

Employee Leave Management System (ELMS)

Project Overview

The **Employee Leave Management System (ELMS)** is a web-based application that helps organizations manage employee leave requests digitally. Employees can apply for leave, track approval status, and view leave balances, while managers and HR administrators can review, approve, reject, and report on leave requests.

This project demonstrates:

- Authentication & Authorization
 - Role-Based Access Control (RBAC)
 - Workflow Management
 - CRUD Operations
 - Reporting & Analytics
 - Email Notifications
 - REST API Development
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Problem Statement

Many small and medium-sized organizations still manage employee leave using emails, spreadsheets, or paper forms. This leads to:

- Manual approval processes
- Difficulty tracking leave balances
- Lack of transparency
- Reporting challenges
- Increased administrative effort

The Employee Leave Management System aims to automate the complete leave request and approval workflow.

Objectives

- Allow employees to apply for leave online.
 - Enable managers to approve/reject leave requests.
 - Track leave balances automatically.
 - Generate leave reports.
 - Maintain leave history.
 - Reduce manual HR work.
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User Roles

Employee

Can:

- Login
- View profile
- Apply for leave
- Cancel pending leave
- View leave history
- Check leave balance

Cannot:

- Approve leaves
 - Manage employees
-

Manager

Can:

- View team leave requests
- Approve leave
- Reject leave
- Add comments

Cannot:

- Manage system settings
-

HR/Admin

Can:

- Manage employees
 - Manage leave types
 - View all leave requests
 - Generate reports
 - Configure leave policies
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Core Modules

1. Authentication Module

Features:

- Login
 - Logout
 - Password Reset
 - Role Management
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2. Employee Management Module

Features:

- Add Employee
- Edit Employee
- View Employee Details
- Deactivate Employee

Fields:

Employee ID

First Name

Last Name
Email
Department
Designation
Joining Date
Manager
Status

3. Leave Type Management

HR can configure leave types.

Examples:

Annual Leave
Sick Leave
Casual Leave
Maternity Leave
Paternity Leave
Work From Home

Fields:

Name
Allowed Days
Carry Forward
Description

4. Leave Request Module

Employee submits leave request.

Fields:

Leave Type
Start Date
End Date
Reason
Attachment

System automatically calculates:

Total Leave Days
Remaining Balance

Workflow:

Employee
↓
Manager Review
↓
Approved / Rejected

5. Leave Approval Module

Manager actions:

Approve
Reject
Add Comment

Status values:

Pending
Approved
Rejected
Cancelled

6. Leave Balance Module

Tracks available leave.

Example:

Annual Leave : 20
Used : 8
Remaining : 12

7. Reports Module

Generate reports:

Employee Report

Employee
Leaves Taken
Remaining Balance

Department Report

Department
Total Leaves
Approved
Rejected

Monthly Report

Month
Leave Statistics

Export:

PDF
CSV
Excel

Database Schema

users

id
name
email
password
role
department_id
manager_id
created_at
updated_at

departments

id
name

description
created_at
updated_at

leave_types

id
name
allowed_days
carry_forward
description
created_at
updated_at

leave_balances

id
user_id
leave_type_id
year
allocated_days
used_days
remaining_days
created_at
updated_at

leave_requests

id
user_id
leave_type_id

start_date
end_date

days_requested

reason

status

manager_comment

approved_by

approved_at

created_at

updated_at

attachments

id

leave_request_id

file_name

file_path

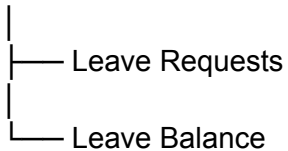
created_at

Entity Relationships

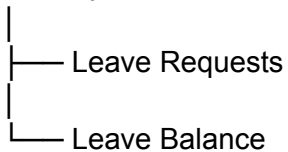
Department



Employee



Leave Type



API Endpoints

Authentication

POST /api/login
POST /api/logout
POST /api/register
GET /api/profile

Employees

GET /api/employees
POST /api/employees
GET /api/employees/{id}
PUT /api/employees/{id}
DELETE /api/employees/{id}

Leave Types

GET /api/leave-types
POST /api/leave-types
PUT /api/leave-types/{id}
DELETE /api/leave-types/{id}

Leave Requests

GET /api/leaves
POST /api/leaves
GET /api/leaves/{id}
PUT /api/leaves/{id}
DELETE /api/leaves/{id}

Approval APIs

POST /api/leaves/{id}/approve

POST /api/leaves/{id}/reject

POST /api/leaves/{id}/cancel

Reports

GET /api/reports/monthly

GET /api/reports/department

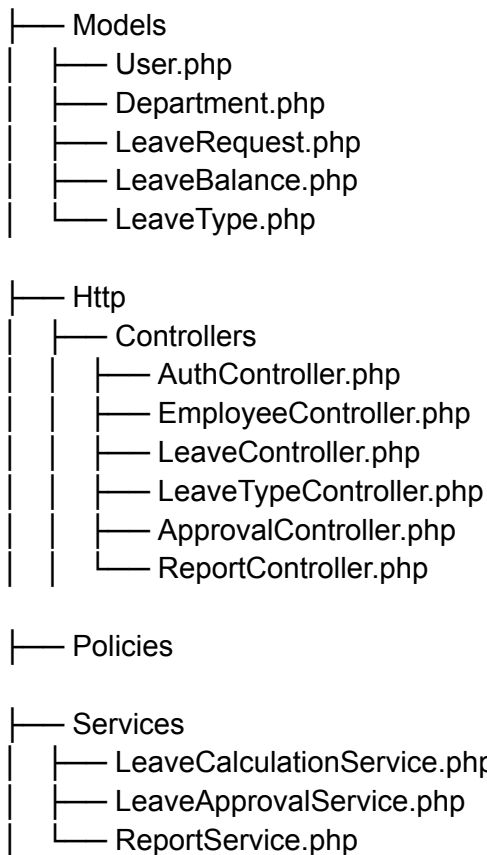
GET /api/reports/employees

GET /api/reports/export/pdf

GET /api/reports/export/csv

Suggested Laravel Folder Structure

app/



15-Day Development Plan

Days 1–2

- Project setup
- Database design
- Migrations

Days 3–4

- Authentication
- User roles

Days 5–6

- Employee management

Days 7–8

- Leave type management

Days 9–10

- Leave request module

Day 11

- Approval workflow

Day 12

- Leave balance calculations

Day 13

- Reports

Day 14

- Email notifications

Day 15

- Testing
- Deployment
- Documentation